

ChE 381 – Chemical Engineering Laboratory I- Fall 2024

Student Learning Outcomes 1, 3 (written and oral), 4, 5, 6, 7

Rubric for assessment for Pre-lab Report, Final Lab Report, SACHE Safety Tutorial, Safety Quiz, Teamwork

RESULTS BY PERFORMANCE INDICATOR: Prelab, Final Lab Reports												
Rubric item (PI)	1	2	3	4	5	6	7	8	9	10	11	12
Mapped SLO	1,3	3,6	3,4,7	3	1,3	1,7	1,3,6	1,3,6	3,7	1,3,6	1,3,6	1,3,6
Average	2.69	2.90	2.90	2.51	2.82	2.79	2.59	2.33	2.59	2.90	2.62	2.90
Std Dev	0.66	0.31	0.31	0.68	0.39	0.41	0.68	0.81	0.68	0.31	0.31	0.31
Average	2.72											

RESULTS BY PERFORMANCE INDICATOR: Oral Presentation				
Rubric item	13	14	15	16
Mapped SLO	3	3	3	3
Average	2.62	3.00	2.56	2.77
Std Dev	0.49	0.00	0.50	0.43

RESULTS BY PERFORMANCE INDICATOR: Safety		
Rubric item	17	18
Mapped SLO	4	4
Average	2.74	2.67
Std Dev	0.64	0.70

RESULTS BY PERFORMANCE INDICATOR: Teamwork			
Rubric item	19	20	21
Mapped SLO	5	5	5
Average	2.82	2.66	2.74
Std Dev	0.25	0.31	0.18

RESULTS BY STUDENT OUTCOME ALL ASSESSMENTS							
	1	3	3	4	5	6	7
		Written	Oral				
Average Score	2.71	2.69	2.74	2.77	2.74	2.71	2.76
Standard dev.	0.19	0.18	0.20	0.12	0.06	0.23	0.16

Average scores showed most students were very strong in developing and conducting experiments, analyzing data, and performing error analysis as well as formulating and solving the complex engineering problems in the lab experiments (SLO 1, 6, 7). Error analysis scores (PI #11) have improved from Fall 2022 and Fall 2023. A large majority were satisfactory in the written lab reports and the final oral presentation (SLO 3). A majority of students had satisfactory knowledge of safety, as assessed by the SACHE tutorial and the assignment on the industrial chemical hygiene plan (SLO 4).

Assessment Rubric for Lab reports (pre-lab and final lab report sections combined)

Report Section; Performance Indicators (PI)	SLO's	(1) Unsatisfactory	(2) Developing	(3) Satisfactory
1. Objective	SLO 1, 3	Main aspects of the lab experiments are not addressed; no experimental objective, experimental description	Several main aspects of lab experiment not addressed or not stated in a clear and concise manner.	Main aspects of lab experiment addressed and clearly stated; minor grammar and usage issues.
2. Experimental: Apparatus, Materials & Supplies, Procedure	SLO 3, 6	Incomplete process diagram, missing materials/supplies list, incomplete or vague experimental procedure	Adequate process diagram, material/ supply list, experimental procedure; several problems or mistakes	Complete process diagram, material/ supply list, experimental procedure with minor problems
3. Project Safety Evaluation	SLO 3, 4, 7	Missing or lacking an understanding of major/ minor safety hazards and safe operating conditions	Safety understanding, missing some major/ minor safety hazards	Safety understanding and important safety hazards, missing some minor hazards
4. Executive Summary	SLO 3	Illegible or incoherent summary of conclusions, poor grammar and formatting	A few problems with summary of conclusions, legibility or neatness	Minor problems with summary of conclusions, grammar, formatting, neatness
5. Introduction	SLO 1, 3	Main aspects of the lab experiments are not addressed; no experimental objective, experimental description or discussion of industrial applications, poor grammar and usage	Several main aspects of lab experiment not addressed or not stated in a clear and concise manner.	Main aspects of lab experiment addressed and clearly stated; minor grammar and usage issues.
6. Theory	SLO 1, 7	Does not appear to grasp the connection between theory and the problem	Some gaps in understanding the application of theory to the problem and expects theory to predict reality	Translates academic theory into engineering applications and accepts limitations of mathematical models of physical reality
7. Results	SLO 1, 3, 6	Drawings, graphs and tables unacceptable, confusing, or missing	Several deficiencies in drawings, graphs and tables. Figures and tables not referred to in text.	Only a few deficiencies with drawings, graphs and tables
8. Discussion	SLO 1, 3, 6	Main points and trends from results are missing or not discussed clearly; lack of understanding of the	Several main points and trends are missing or not discussed clearly;	Only a few points or trends are missing or not discussed clearly; a substantial

		engineering reasoning behind the analysis	some understanding of engineering reasoning behind analysis	understanding of the engineering reasoning behind analysis
9. References	SLO 3, 7	Seeks no extra information other than what is provided by instructor	Seeks information from a few sources, mainly from the textbook or instructor	Seeks information on problems from multiple sources
10. Data Tabulation	SLO 1, 3, 6	Experimental data tables and graphs unacceptable or missing	Several deficiencies with data tables and graphs	Only a few deficiencies in data tables and graphs
11. Sample Calculations	SLO 1, 3, 6	Calculations are incorrect, presented in a confusing way and do not link with specific experimental results	Some calculations are incorrect or presented in a confusing way	One or two calculations are incorrect or presented in a confusing way
12. Error Analysis	SLO 1, 3, 6	Cursory and qualitative description of errors	Qualitative analysis of error with some calculations	Quantitative and qualitative error analysis, missing some minor calculations

Assessment Grading for Lab reports (pre-lab and final lab report sections combined)

Report Section	SLO's	(1) Unsatisfactory	(2) Developing	(3) Satisfactory
1. Objective	SLO 1, 3	0-1	2-3	4-5
2. Experimental: Apparatus, Materials & Supplies, Procedure	SLO 3, 6	0-11	12-15	16-20
3. Project Safety Evaluation	SLO 3, 4, 7	0-5	6-7	8-10
4. Executive Summary	SLO 3	0-5	6-7	8-10
5. Introduction	SLO 1, 3	0-5	6-7	8-10
6. Theory	SLO 1, 7	0-5	6-7	8-10
7. Results	SLO 1, 3, 6	0-8	9-11	12-15
8. Discussion	SLO 1, 3, 6	0-8	9-11	12-15
9. References	SLO 3, 7	0-5	6-7	8-10
10. Data Tabulation	SLO 1, 3, 6	0-5	6-7	8-10
11. Sample Calculations	SLO 1, 3, 6	0-5	6-7	8-10
12. Error Analysis	SLO 1, 3, 6	0-5	6-7	8-10

Assessment Rubric for Oral Presentation

Performance Indicator (PI)	SLO's	(1) Unsatisfactory	(2) Developing	(3) Satisfactory
13. Organization	SLO 3	Talk is poorly organized, e.g. no clear introduction or summary of talk is presented; time is too short or too long	. Presents key elements of an oral presentation adequately, but "tell them" not clearly applied; time is slightly too long or too short	Plans and delivers an oral presentation effectively; applies the principle of "(tell them)3" --well organized; time is just right
14. Delivery	SLO 3	"Major difficulties with the mechanical aspects of the presentation · No eye contact · Difficult to hear or understand speaking · Reads from prepared script · Blocks the screen · Distracting nervous habits (um, ah, clicking pointer, etc.)"	"Has some minor difficulties with the mechanical aspects of the presentation · Eye contact is sporadic · Occasionally difficult to hear or understand speaking · Overuses prompts or does not use prompts enough- occasionally stumbles or loses place; appears to have memorized presentation	Presents well mechanically: · Makes eye contact · Can be easily heard · Speaks comfortably with minimal prompts (notecards) · Does not block screen · No distracting nervous habits
15. Visual Aids	SLO 3	Multiple slides are unclear or incomprehensible	Visual aides have minor errors or are not always clearly visible	Uses visual aides effectively
16. Content	SLO 3	Omits key results during presentation	Presentation is lacking some details or technical content	Presentation has enough detail appropriate and technical content for the time constraint and the audience

Assessment Rubric for Lab Teams/Teamwork

Performance Indicator (PI)	SLO's	(1) Unsatisfactory	(2) Developing	(3) Satisfactory
17. Pre-Lab Teamwork	SLO 5	Does not contribute to work or absent from lab sessions >50% of the time. Hides in the background; only participates if strongly encouraged	Sometimes depends on others to complete the work; contributes less than fair share Takes charge when not in the position to lead and meets the objectives of the role.	Contributes a fair share to the project workload. Demonstrates the ability to assume a designated role in the group and able to meet the objectives of these roles
18. Final Lab Teamwork	SLO 5	Does not contribute to work or absent from lab sessions >50% of the time. Hides in the background; only participates if strongly encouraged.	Sometimes depends on others to complete the work; contributes less than fair share. Takes charge when not in the position to lead and meets the objectives of the role.	Contributes a fair share to the project workload. Demonstrates the ability to assume a designated role in the group and able to meet the objectives of these roles
19. Oral Presentation Teamwork	SLO 5	Does not contribute to presentation or absent from presentation. Hides in the background; only participates if strongly encouraged	Sometimes depends on others to complete the presentation, or answer questions during presentation	Contributes a fair share to the presentation.

Oral Presentation Evaluation Form

CHE 381 FINAL LAB PRESENTATION EVALUATION FORM

TEAM #: _____ LAB EXPERIMENT: _____

RUBRIC: _____ SCORE: _____

1. (50 POINTS) Technical Content and Understanding:

- | | |
|--|----------|
| a. Introduction (Objective/ Industry apps) | (10 pts) |
| b. Theory | (5 pts) |
| c. Experimental | (5 pts) |
| d. Results | (10 pts) |
| e. Analysis & Discussion | (10 pts) |
| f. Conclusions | (10 pts) |

COMMENTS:

SCORE: _____

2. (20 POINTS) Visual Aids

- | | |
|--|----------|
| a. Clarity of graphs/tables/figures | (10 pts) |
| b. Use of space on slides | (10 pts) |
| c. Extras (nice graphics, animations, etc) | (Bonus) |

COMMENTS:

SCORE: _____

3. (10 POINTS) Use of Time, 10 minutes $\pm$ 1 minute)

- | | |
|---------------|----------------|
| a. Too long | (-1 to -2 pts) |
| b. Too short | (-1 to -2 pts) |
| c. Just right | (10 pts) |

COMMENTS:

SCORE: _____

4. (20 POINTS) Presentation Delivery

- | | |
|------------------------|---------|
| a. Eye contact | (8 pts) |
| b. Voice level | (7 pts) |
| c. Answering questions | (5 pts) |

COMMENTS:

PRESENTATION GRADE: (100 points max): _____

Assessment Rubric for Safety Assignments

Performance Indicator (PI)	SLO's	(1) Unsatisfactory	(2) Developing	(3) Satisfactory
Understanding industrial safety and practices	SLO 4	Missing or lacking an understanding of industrial safety practices	Has some understanding of industrial hazards and safety practices	Has a good understanding of industrial hazards and safety practices Example: Passed a safety tutorial (SACHE)
Understanding of lab safety and safety evaluations	SLO 4	Missing or lacking an understanding of major/ minor safety hazards and safe operating conditions	Missing some major/minor aspects of laboratory safety practices	Satisfactory understanding of laboratory safety covered. Example: Understands a laboratory Chemical Hygiene Plan

Safety Assignment: SACHE Tutorial, "Chemical Reactivity Hazards"

SACHE-AICHE safety tutorial website

AICHE (American Institute of Chemical Engineers) has created a website with various safety tutorials covering process safety, risk assessment and chemical reactivity hazards.

ASSIGNMENT: Complete the tutorial titled "Chemical Reactivity Hazards" in the website. Given below is the procedure on how to access the website.

NOTE: You will be asked for either your AICHE member ID or an AICHE user ID. If you do not have either one, you have the option of joining AICHE as a student member (free) or signing up for access to the AICHE website. I SUGGEST YOU DO THIS EARLY IN THE SEMESTER IN CASE THERE ARE PROBLEMS WITH SIGNING IN TO THE WEBSITE. THERE WILL BE NO EXTENSIONS FOR LATE COMPLETIONS.

GRADING: You must email proof (Certificate issued by SACHE) that you have completed the tutorial and answered the questions for full credit.

DUE: Wednesday November 26, 2014 at 1:00 PM

WEBSITE PROCEDURE:

1. Go to www.sache.org and click on "Login" in the Menu on the left hand side of the page.
2. Use the Alternate Login: University Student Login
 - a. Select University: University of Illinois at Chicago
 - b. Type the Password: chemeisgreat (no cap or spaces)
3. Click on "Student Certificate Program" to get into the tutorial section. Pick the safety tutorial:
 - a. Chemical Reactivity Hazard

You will then need to type in your AICHE member ID/password or AICHE user ID /password in order to gain access to the specific tutorial. If you do not have either one- sign up to be a student member or to be a AICHE website user.

4. Print out Certificate that you completed the tutorial and email to TA for credit.

Good Luck!

Professor Zdunek

Safety Assignment: Industrial Chemical Hygiene Plan

A Chemical Hygiene Plan (CHP) is an OSHA required safety document at all R&D facilities. A CHP contains useful information on chemical and gas handling, conducting experiments safely in a laboratory setting, employee responsibilities, as well as what to do in an emergency. Employees are required to read the document annually and sign a release form.

ASSIGNMENT: A CHP and 1-page Safety Rules document from a corporate R&D Center have been provided in this Assignment folder (also located in the Safety folder). Please review the documents. You will be responsible for material in the following Chapters: Chapter 1, 3, 2, 4, 5, 6, 7, 8. and the Definitions at the End. You can skip Chapter 2, 3.1 and 3.3 and the Appendices and Attachments.

TAKE THE SAFETY QUIZ: Due Tuesday September 5th at end of lab period. Be brief but concise in your answers. Open book/open notes and team discussion is allowed. Email or provide a hard copy of your completed quiz to the TA's for grading. If you have any questions let me or the TA know.

Note: If you download the CHP file, make sure you download the image3.gif file as well

SAFETY QUIZ: CHE381- Chemical Engineering Laboratory I

Please answer the following questions testing your knowledge of the content of the Chemical Hygiene Plan safety document. Be brief but concise in your answers. Open book/open notes and team discussion is allowed but answers must be in your own words.

1. What is the most fundamental rule of chemical safety?

Awareness is the single most fundamental rule of chemical safety

2. What two pieces of personal protective equipment must be worn at all times in the lab?

Safety glasses and safety shoes

3. What additional kinds of personal protective equipment are available depending on the chemical hazard or experimental conditions?

Lab coats, face shields, gloves, splash aprons, ear protection

4. True or False: Fume hoods are generally used as storage areas for chemicals.

False, they are used for conducting experiments and temporary storage of chemicals.

5. Fume hoods are to be used when the Threshold Limit Value (TLV) of a chemical is at or below:

TLV of less than 1000 ppm

6. What are the four routes of chemical entry into the body?

Inhalation, ingestion, injection, eye/skin contact

7. True or False: Respirators are generally used in the R&D laboratories for normal work procedures.

False, respirators are generally not used for normal procedure, periodic use is OK

8. What is the role of the Chemical Hygiene Plan?

This Chemical Hygiene Plan describes American Air Liquide's safety program, including (but not limited to) chemical safety rules, personal protective equipment, control equipment (i.e., fume hoods), employee training programs, and medical programs. The Chemical Hygiene Plan is designed as a tool to coordinate safety procedures.

9. What is the guideline for laboratory room airflow and fume hood airflow?

Laboratory airflow: 20 linear feet per minute, Fume hood: 80-120 linear feet per minute

10. List 3 Common Housekeeping Practices that promote chemical hygiene and safety:

Full List (pick 3):

- **Keep all aisles, hallways, and stairs clear of all chemicals.**
- **Keep all work areas and especially workbenches clear of clutter and obstructions.**
- **All working surfaces and floors should be cleaned regularly.**
- **Access to emergency equipment, showers, eyewashes and exits should never be blocked by anything.**
- **Wastes should be properly labeled, and kept in the proper containers.**
- **Any unlabeled containers are considered wastes by the end of each workday.**

11. What is the proper method of transporting gas cylinders in the laboratory?

Gas cylinders must be transported using a 1-cylinder 4-wheel cart.

12. True or False: MSDS sheets are required only for toxic chemicals and gases used in the laboratory.

False, MSDS sheets are required for all chemicals and gases used in the laboratory.

13. What is an accepted method of vapor detection in the laboratory?

Accepted methods of vapor detection; safety monitors, detector tubes (Draeger tubes), detection meters. Odor detection is not an accepted method.

14. Name 3 Personal Hygiene Practices that should be followed in the laboratory.

Pick 3:

- **Wash promptly if skin contact is made with any chemical, regardless of corrosivity.**
- **Wear appropriate eye protection at all times.**
- **Avoid inhalation of chemicals, do not "sniff" test chemicals.**
- **Do not mouth pipette anything, use suction bulbs.**
- **Wash well before leaving the laboratory; do not wash with solvents, use soap.**
- **Change clothing as soon as possible after leaving laboratory, and launder work clothes often.**
- **Do not eat or drink in chemical areas.**
- **Do not bring food, beverage or tobacco products into chemical storage or use areas.**

15. Describe how individual chemicals of less than 15 lbs. (for example a bottle of acetone) are safely transported.

Individual chemicals in a container are transported using rubber "bucket-type" carriers.

16. List the Emergency Procedure to be followed for a large chemical spill or toxic gas release in the laboratory.

- **Evacuate the immediate area.**
- **Dial "0" (Facility Operator), advise nature of emergency, and location of emergency (e.g., "Chemical Spill, XYZ lab").**
- **The Operator will notify all employees (on the Public Address system) to evacuate, and will activate the Emergency Response Team.**
- **Immediately exit the lab and deactivate the lab power by pushing the emergency (red) power button outside of the lab.**
- **Evacuate the building through the nearest and safest exit.**
- **Unless otherwise notified (on the P.A. system), assemble in front of the facility at the middle driveway.**
- **Report to the department secretary**

17. What is the difference between a flammable chemical and a combustible chemical in terms of OSHA and NFPA guidelines.

"Flammable" is generally used to refer to chemicals with a flash point below 100° F. Chemicals with flash points between 100 and 200° F are termed "Combustible".

18. List the Emergency Procedure for an individual chemical exposure in the laboratory.

In the case of a chemical exposure (of a worker), the following steps should be observed:

- **Dial "0" (Facility Operator), advise nature of emergency, and location of emergency**
- **(E.g., "Person with severe chemical burns, ABC lab").**
- **Operator calls Fire Department Paramedics and notifies Emergency Response Team.**
- **If there are no apparent dangers to yourself of chemical exposure, the following may be performed until the E.R.T./Paramedics arrive:**
- **If a person has been splashed with a chemical, wash them with plenty of water for at least 15 minutes, removing all contaminated clothing (Eyewashes for eye/face splashes, Emergency Shower for all other).**
- **If a person has been overexposed by inhalation, get victim to fresh air (if it is safe for you to enter area), apply artificial respiration if necessary.**

19. What is the definition of a "hazardous chemical"?

A chemical for which there is statistically significant evidence that an acute or chronic health effect may occur in exposed employees.

20. All laboratories that handle chemicals must be equipped with eyewashes and safety showers. What is the recommended flowrate of this equipment and how often should they be checked?

Safety shower: 30 gallons per minute. Eye showers: 3 gallons per minute. Both checked monthly.